Practice workbook · Draw, label and compare the structure of a neurone

Drawing a labelled motor neurone, naming a part such as the axon, giving the role of the myelin sheath, and spotting structural differences between two neurone drawings.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DRAW-LABEL-NEURONE-PARTS

1. Question 3 · part a.i · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part a.i only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

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(ii) Describe the function of a motor neurone.

(2)

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1. Question 3 · part a.i · 4 marks · Working and answer

Working and answer

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2. Question 7 · part b.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

7 (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

(1)

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

(1)

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

(2)

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2. Question 7 · part b.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

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(ii) A motor neurone is a specialised cell.

Draw a diagram of a motor neurone.

Your diagram should include these labels.

- nucleus
- cell body
- myelin sheath

(4)

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(iii) Describe the function of a motor neurone in the nervous system.

(2)

2. Question 7 · part b.ii · 4 marks · Working and answer

Working and answer

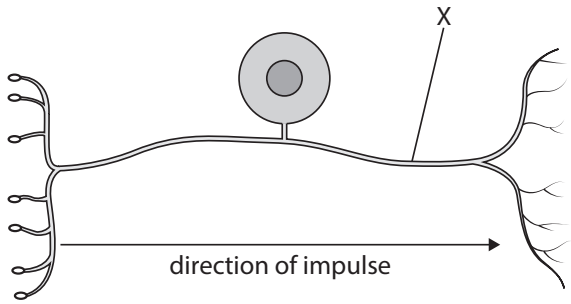
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3. Question 1 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

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(ii) Name the structure labelled X.

(1)

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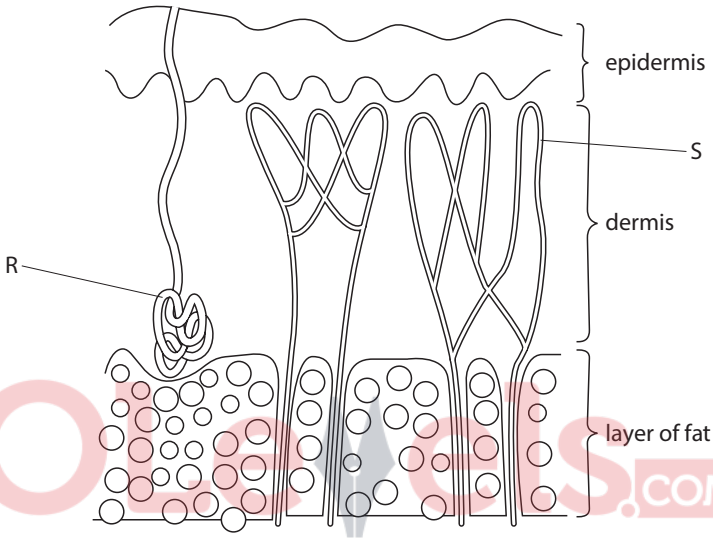
3. Question 1 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

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part S

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(ii) Describe how part R helps to cool the body.

(2)

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3. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

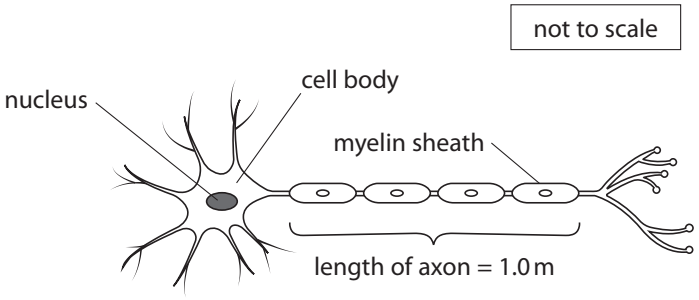
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4. Question 5 · part a.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part a.ii only.

5 The diagram shows a motor neurone.



(a) (i) The actual diameter of the nucleus of the motor neurone is 12 µm.

A student observes the motor neurone using a ×50 microscope lens.

Calculate the diameter of the image seen after magnification.

Give your answer in millimetres.

(2)

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diameter = ... mm

(ii) Neurones transmit electrical impulses from one part of the body to another.

Suggest one function of the myelin sheath surrounding the axon.

(1)

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(b) Nerve impulses travel at an average speed of 1.2×10^8 µm per second.

Hormones travel at an average speed of 1.5×10^6 µm per second.

(i) Explain this difference in speed.

(2)

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4. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 5 · part c.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.i only.

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5 This question is about the nervous system.

(a) Describe what is meant by a reflex action.

(3)

(b) Diagram 1 shows a reflex arc.

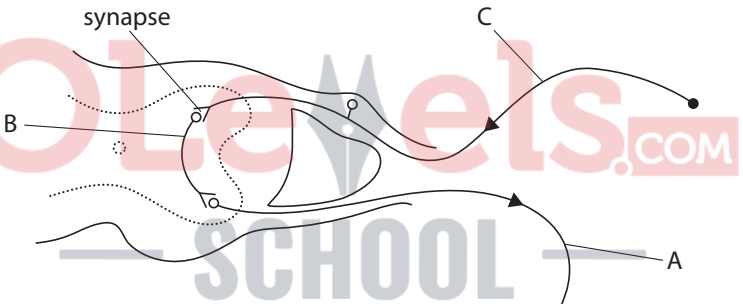


Diagram 1

Name the types of neurone labelled A, B and C.

(3)

A .....

B .....

C .....

5. Question 5 · part c.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.i only.

(c) Diagram 2 shows a neurone from a healthy brain and a neurone from the brain of someone with Alzheimer's disease.

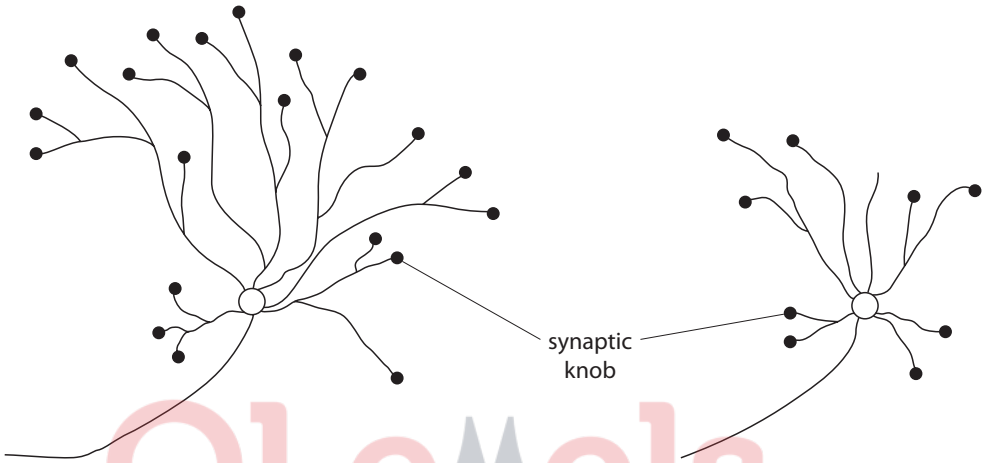


Diagram 2

(i) Give three differences between the two neurones. **(3)**

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2 .....

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3 .....

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5. Question 5 · part c.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.i only.

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(ii) Acetylcholine is a neurotransmitter.

Explain the importance of acetylcholine in the functioning of the brain.

(3)

(iii) One symptom of Alzheimer's disease is that less acetylcholine is produced in the brain.

Explain why there is less acetylcholine produced in people with Alzheimer's disease.

Use information from diagram 2 to help your answer.

(3)

(Total for Question 5 = 15 marks)

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5. Question 5 · part c.i · 3 marks · Working and answer

Working and answer

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